

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Final
Duration: 90 minutes

Semester: Fall 2025
Full Marks: 45

CSE 440: Natural Language Processing II

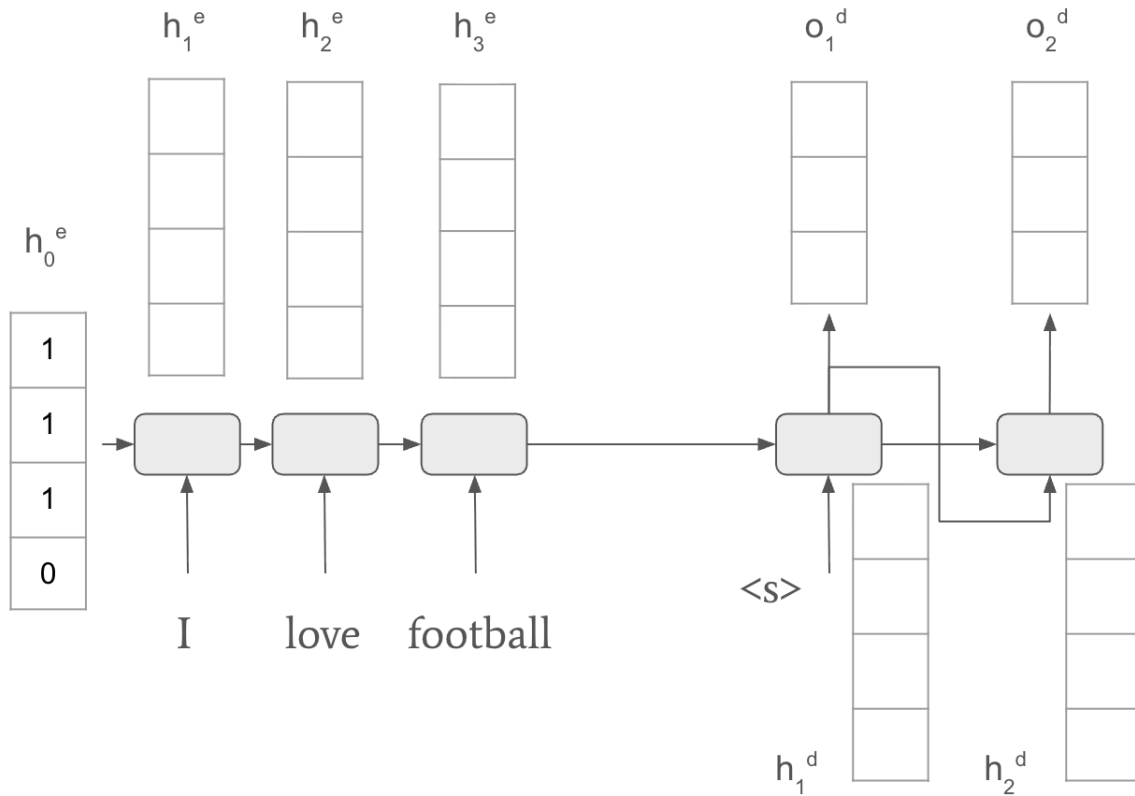
SET A

Name:	ID:	Section:
Total marks		

You have to answer all of the questions. This question is your answer script– write your answers where indicated. Use the provided answer sheet to do any rough calculations you need to do. Do not write your rough calculations on this sheet.

1.

[20]



Fill out the empty vectors. This is attention RNN. All the values and equations you will need are given below. Calculate up to 3 decimal points.

$$h_0^e = [1 \ 1 \ 1 \ 0]^T$$

$$h_0^d = h_3^e$$

$$W^e = [0 \ 0 \ 0, -1 \ 0 \ 0, 1 \ -1 \ 1, -1 \ 0 \ -1]$$

$$U^e = [1 \ -1 \ 1 \ -1, 0 \ 0 \ 0 \ 0, 1 \ 0 \ 1 \ 0, 0 \ 1 \ 0 \ 1]$$

$$W^d = [0 \ 0 \ 0, 1 \ 0 \ 0, -1 \ -1 \ 1, -1 \ 0 \ 1]$$

$$U^d = [1 \ 1 \ 0 \ -1, 0 \ 0 \ 0 \ 0, 1 \ 1 \ -1 \ 0, 0 \ 0 \ 0 \ 1]$$

$$V^d = [-1 \ 1 \ 0 \ 0, 0 \ 0 \ 0 \ 0, 1 \ -1 \ 1 \ 0]$$

$$\langle s \rangle = [0 \ 0 \ 0]^T = \hat{y}_0$$

$$I = [0 \ 1 \ 0]^T = x_1$$

$$\text{love} = [0 \ 0 \ 1]^T = x_2$$

$$\text{football} = [1 \ 0 \ 0]^T = x_3$$

$$h_j^e = W^e x_j + U^e h_{j-1}^e$$

$$h_i^d = W^d \hat{y}_{i-1} + U^d h_{i-1}^d + c_i$$

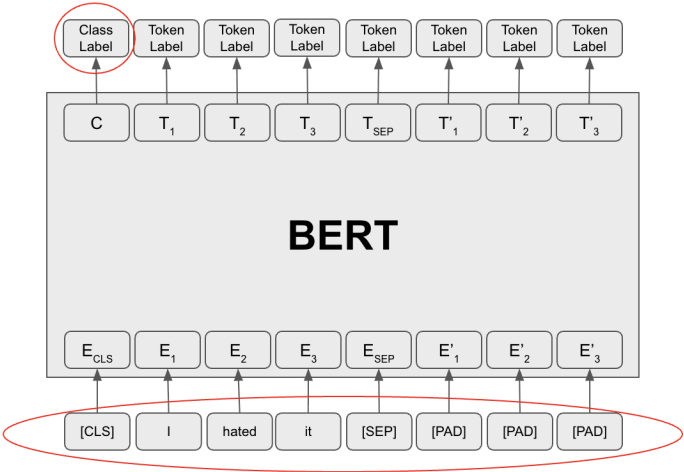
$$\alpha_{ij} = \text{softmax}(\text{score}(h_{i-1}^d, h_j^e)) \text{ for all } i \text{ and } j$$

$$\text{score}(h_i^d, h_j^e) = h_i^d \cdot h_j^e \text{ for all } i \text{ and } j$$

$$c_i = \sum \alpha_{ij} h_j^e \text{ for all } i \text{ and } j$$

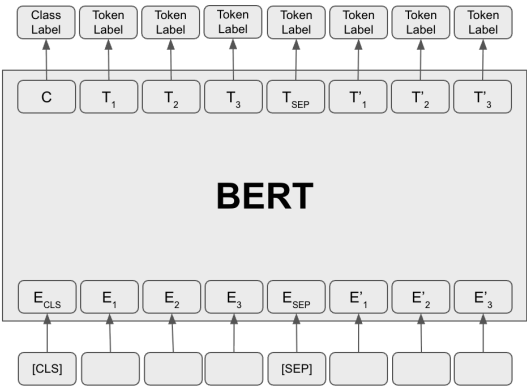
$$o_i^d = \text{softmax}(V^d h_i^d) = \hat{y}_i$$

2. For each of the following tasks, indicate what should go into the input into BERT, and which output should be considered. Let me give you an example: if I want to do a sentiment classification on the sentence “I hated it”, this should be my answer:

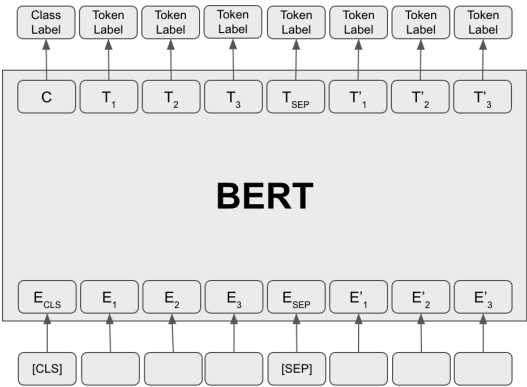


Now do this for the following ones:

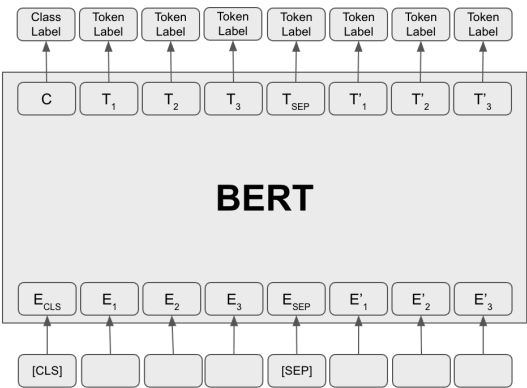
Task: PoS tagging
Sentence: I hated it



Task: NLI
Sentence 1: I saw it
Sentence 2: I hated it



Task: NER
Sentence: Rayhan won today



3. Each question is worth 5. Use this page and the next one to answer.

[15]

- a. What are the three approximations we studied for a Hidden Markov model PoS tagger?
- b. How do you calculate tag transition probability and emission probability from an existing annotated corpus (e.g. Brown Corpus)?
- c. Write three disadvantages of a Hidden Markov Model PoS tagger.